VICTORIA UNIVERSITY
MELBOURNE AUSTRALIA

NIT3171 ICT Business Analytics and Data Visualization
Diabetes Dataset Analysis Prediction
Individual Project

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Table of Contents

List of Figures

List of Tables

Introduction

Diabetes is a non-communicable disease that is chronic and arises when the body fails to properly control the blood sugar levels. The condition occurs as a result of insulin production or the inability of the body to utilize insulin. Serious health issues that may arise due to uncontrolled diabetes overtime include heart disease, kidney failure, vision loss and nerve damage. Because of its rising rates across the world and the long term health outcome of diabetes, it is one of the leading concerns of the public health and, therefore the detection and risk evaluation of this disease at an early age cannot be underrated.

1. Dataset Selection

The dataset "Diabetes" was chosen to be analyzed. The dataset has gained significant popularity in academic and educational studies of risk factors of diabetes and the creation of predictive machine learning models. Reliability, transparency and reproducibility of the analysis are guaranteed by using a publicly accessible and well documented dataset.

This dataset was selected based on its structured format, manageable size as well as its applicability to the real life healthcare analytics. It allows the exploration of data, visualization, preprocessing, and assessment of the models effectively in the framework of the project. Moreover, the extensive applicability of the dataset in the previous studies proves its validity and accuracy of data.

According to the assessment guidelines, the dataset was forwarded to the lecturer before the assignment began to ensure that it was confirmed and approved. The lecturer checked the dataset and approved it

Source: Kaggle

Variable: 09

- Demographic information - Age and number of Pregnancies.
- Clinical measurements - Glucose, Blood Pressure, Skin Thickness, Insulin level, and Body Mass Index (BMI).
- Genetic factor - Diabetes Pedigree Function (a score estimating hereditary risk based on family history).
- Diabetes-related outcome - Outcome show as binary classification 1 for positive diabetes diagnosis, 0 for negative it means non diabetic

Records: 768

Validation: Validated by module Lecturer Mrs. Yasanthika

Limitations:

- There are missing values that need to be handled carefully.
- Numbers of records have the value 0 of such as glucose, blood pressure, skin thickness, insulin, BMI.
- The Outcome variable is imbalanced, with 500 non diabetes cases and 268 diabetes cases .

Tools: I selected Python with required libraries instead of WEKA due to its flexibility and it being able to support more advanced data visualization and detailed model evaluation.

2. Data Preprocessing

I. **Handle missing values**

- In this data set, a number of records have the value 0 of such attributes as glucose level, blood pressure, insulin, and BMI. From a medical perspective, these attributes cannot realistically be equal to zero for a living person. Therefore, these zero values do not represent actual measurements; instead, they indicate missing or unrecorded data. As a result, these zero entries are treated as missing values during the data preprocessing stage.

II. **Encoding Categorical Variables**

- None of the columns in this data are categorical, string, or text-based. All features are numerical, therefore there is no need to encode.

III. **Check for Duplicate Rows**

- There is no duplicate rows

IV. **Outlier Detection**

V. **Handling class imbalance**

- I identified that there are 500 non diabetes cases and 268 diabetes cases. This step is done using the technique SMOTE.

```python
#Check the number of Missing Values
columns_with_invalid_zeros = ['Glucose', 'BloodPressure', 'Insulin', 'BMI']
df[columns_with_invalid_zeros].eq(0).sum()
```
✓ 0.0s Python

```
Glucose            5
BloodPressure     35
Insulin          374
BMI               11
dtype: int64
```

Figure 1 Identify the number of missing values

There are 5 missing values in Glucose column, 35 missing values in BloodPressure, Insulin attribute contain 374 missing values and 11 missing values in BMI attribute.

```python
# List of columns where 0 is not valid
zero_not_valid = ['Glucose', 'BloodPressure', 'Insulin', 'BMI']

# Replace 0 with NaN
df[zero_not_valid] = df[zero_not_valid].replace(0, np.nan)
```
✓ 0.0s Python

```python
#Recheck missing values
df.isnull().sum()
```
✓ 0.0s Python

```
Pregnancies                   0
Glucose                       5
BloodPressure                35
SkinThickness                 0
Insulin                     374
BMI                          11
DiabetesPedigreeFunction      0
Age                           0
Outcome                       0
dtype: int64
```

Figure 2 Replace missing values with NaN

```
# Impute numerical columns with median
for col in zero_not_valid:
    df[col].fillna(df[col].median(), inplace=True)
```

[203] 0.0s Python

```
C:\Users\BC\AppData\Local\Temp\ipykernel_6440\4077674720.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace
The behavior will change in pandas 3.0. This inplace method will never work because the intermediate object on which we are setting values always behaves as a copy.

For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inp


  df[col].fillna(df[col].median(), inplace=True)
C:\Users\BC\AppData\Local\Temp\ipykernel_6440\4077674720.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace
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C:\Users\BC\AppData\Local\Temp\ipykernel_6440\4077674720.py:3: FutureWarning: A value is trying to be set on a copy of a DataFrame or Series through chained assignment using an inplace
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For example, when doing 'df[col].method(value, inplace=True)', try using 'df.method({col: value}, inplace=True)' or df[col] = df[col].method(value) instead, to perform the operation inp


  df[col].fillna(df[col].median(), inplace=True)
```

Figure 3 Impute missing values

```
#Recheck After Imputation
df.isnull().sum()
```

0.0s Python

```
Pregnancies                 0
Glucose                     0
BloodPressure               0
SkinThickness               0
Insulin                     0
BMI                         0
DiabetesPedigreeFunction    0
Age                         0
Outcome                     0
dtype: int64
```

Figure 4 Recheck missing values after imputation

```
#Check for duplicate rows
df.duplicated().sum()
```

0.0s Python

```
np.int64(0)
```

Figure 5 Check for duplicate rows

```python
# Check class distribution
print(df['Outcome'].value_counts())
```

✓ 0.0s Python

```
Outcome
0    500
1    268
Name: count, dtype: int64
```

```python
# Check percentage distribution
print(df['Outcome'].value_counts(normalize=True) * 100)
```

✓ 0.0s Python

```
Outcome
0    65.104167
1    34.895833
Name: proportion, dtype: float64
```

Figure 6 Identify the class imbalance

```
Before SMOTE:
Outcome
0    400
1    214
Name: count, dtype: int64
After SMOTE:
Outcome
0    400
1    400
Name: count, dtype: int64
```

Figure 7 After handling the class imbalance

3. Literature Review

Recent research studies regarding clinical indicators of glucose, insulin, BMI, age, and genetic predisposition were related to enhance understanding of present knowledge and gaps on diabetes risk factors, prevalence, and predictive modelling. These studies can be used to formulate the research questions of this project by establishing the important variables related to Type 2 Diabetes.

3.1 Literature Survey

Study 1: Predictive Modeling for Diabetes Using Machine Learning Techniques [5]

- Summary: This paper used machine learning algorithms that included Logistic Regression, Decision Trees, Random Forest and Support Vector Machines to predict diabetes with the

Pima Indians Diabetes data. Some of the clinical variables that were studied included glucose, BMI, insulin, age, and pregnancies.

- Key Findings: The findings revealed that the most predictive factor of diabetes was the level of glucose. Another good contributing factor was BMI and age. Random Forest was accurate than Logistic Regression.

- Why It Matters: As the chosen dataset has the same clinical variables, this paper justifies the research of which variables most impact the prediction of diabetes and the comparison of the models.

Study 2: Association between Obesity, BMI, and Type 2 Diabetes Risk [2]

- Summary: This study looked at how Body Mass Index (BMI) has been associated with the occurrence of Type 2 Diabetes. In the research, a statistical analysis was conducted to determine the effect of obesity on insulin resistance.

- Key Findings: Greater BMI was highly correlated with high risk of diabetes. Obesity causes insulin resistance that raises the levels of blood glucose and causes diabetes diagnosis.

- Why It Matters: The chosen dataset comprises of the BMI and insulin measurements. Thus, the issue can be explored regarding the positive results of diabetes in the data of higher BMI values.

Study 3: Role of Genetic Predisposition in Type 2 Diabetes [3]

- Summary: In this study the risk factors of diabetes examined were the family history and genetic predisposition. It tested the interaction of hereditary factors and insulin production and glucose regulation.

- Key Findings: People who had diabetes in their family history were more probable to acquire Type 2 Diabetes. Insulin resistance and glucose metabolism were also genetic risk factors.

- Why It Matters: The data comprises of DiabetesPedigreeFunction, which signifies hereditary effect. This enables the researcher to investigate the relationship between increased pedigree function values and increased risk of diabetes.

Study 4: Age and Glucose as Risk Factors for Type 2 Diabetes [4]

- Summary: This study examined the dependence of age and levels of fasting glucose with the prevalence of diabetes. Regression analysis was applied in the study to estimate the strength of association.

- Key Findings: The elderly were identified to be at risk of diabetes. The most powerful clinical diagnosis indicator was high levels of fasting glucose.

- Why It Matters: The variables covered in the dataset are Age and Glucose. Thus, the evaluation of their connection with the diagnosis of diabetes can verify the existing research.

3.2 Identified Research gaps

Based on the literature review, the research gaps were as follows:

1. A lot of research is done on each of the factors individually for example BMI or glucose, but little research has been carried out on all of the clinical variables at once with machine learning models.
2. Although glucose has been found as the most significant factor and the interaction of insulin, BMI, and hereditary risk needs to be analyzed further.
3. Certain research primarily make statistical analysis instead of comparisons between different machine learning models to provide more effective prediction.
4. The difference between the importance of features in simpler models (Logistic Regression) and the more complicated models (Random Forest) is not discussed extensively.

3.3 Research Questions

According to the literature review and the variables that were available in the dataset, the following research questions are developed:

1. What are the greatest clinical risk factors that relate to a diagnosis of diabetes?
2. Does Age and diabetes status correlate significantly?

3. What is the strength of the relationship between Glucose and diabetes prediction and other variables?
4. Does an increase in BMI lead to a high probability of diabetes?
5. Does DiabetesPedigreeFunction (hereditary risk) increase the prevalence of diabetes?
6. What machine learning model gives the most effective performance in prediction of diabetes?

The existence of these gaps shows that comprehensive machine learning analysis with utilization of various clinical features is necessary.

Table 1 Overview of selected Research Papers

Paper ID	Authors	Year	Methodology	Main Focus	Key Finding
1	Smith et al.	2023	Machine Learning Models	Clinical prediction	Glucose strongest predictor
2	Johnson & Lee	2022	Statistical Analysis	BMI & Obesity	High BMI increases risk
3	Wang et al.	2021	Genetic Study	Hereditary Risk	Family history increases risk
4	Patel et al.	2023	Regression Analysis	Age & Glucose	Older age increases risk

4. Data Exploration

4.1 Load and examine the dataset

Dataset Structure:

- Number of columns: 09
- Number of rows : 768

```
#Dataset shape (rows, columns)
df.shape
```
✓ 0.0s

```
(768, 9)
```

```
df.head()
```
✓ 0.1s

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
0	6	148	72	35	0	33.6	0.627	50	1
1	1	85	66	29	0	26.6	0.351	31	0
2	8	183	64	0	0	23.3	0.672	32	1
3	1	89	66	23	94	28.1	0.167	21	0
4	0	137	40	35	168	43.1	2.288	33	1

Figure 8 Dataset before preprocessing

Variable Types:

Table 2 Variable, variable type, data type

	Variable	Variable Type	Data type	Description
Demographic information	Age	Numeric	Int64	Age in years
	Pregnancies	Numeric	Int64	Number of times pregnant
Clinical measurements	Glucose level	Numeric	Int64	Plasma glucose concentration
	Blood Pressure	Numeric	Int64	
	Skin Thickness	Numeric	Int64	
	Insulin level	Numeric	Int64	2 hour serum insulin
	BMI	Numeric	float64	Body Mass Index
Genetic factors	Diabetespedigree Function	Numeric	Float64	function of family history and genetics
Target variable	Outcome	Numeric	Int64	0 - no diabetes, 1 - diabetes

```
#Column names and data types
df.info()
✓ 0.0s

<class 'pandas.core.frame.DataFrame'>
RangeIndex: 768 entries, 0 to 767
Data columns (total 9 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   Pregnancies               768 non-null    int64
 1   Glucose                   768 non-null    int64
 2   BloodPressure             768 non-null    int64
 3   SkinThickness             768 non-null    int64
 4   Insulin                   768 non-null    int64
 5   BMI                       768 non-null    float64
 6   DiabetesPedigreeFunction  768 non-null    float64
 7   Age                       768 non-null    int64
 8   Outcome                   768 non-null    int64
dtypes: float64(2), int64(7)
memory usage: 54.1 KB
```

Figure 9 Data types

4.2 Variable Distribution

Summary Statistics:

```
#Summary Statistics
df.describe()
```

✓ 0.0s

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
count	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000	768.000000
mean	3.845052	121.656250	72.386719	20.536458	140.671875	32.455208	0.471876	33.240885	0.348958
std	3.369578	30.438286	12.096642	15.952218	86.383060	6.875177	0.331329	11.760232	0.476951
min	0.000000	44.000000	24.000000	0.000000	14.000000	18.200000	0.078000	21.000000	0.000000
25%	1.000000	99.750000	64.000000	0.000000	121.500000	27.500000	0.243750	24.000000	0.000000
50%	3.000000	117.000000	72.000000	23.000000	125.000000	32.300000	0.372500	29.000000	0.000000
75%	6.000000	140.250000	80.000000	32.000000	127.250000	36.600000	0.626250	41.000000	1.000000
max	17.000000	199.000000	122.000000	99.000000	846.000000	67.100000	2.420000	81.000000	1.000000

Table 3 Summary statistics of the variables

Variable Name	Average	Median	Spread
Pregnancies	3.85	3.00	3.37
Glucose	121.66	117.00	30.44
BloodPressure	72.39	72.00	12.10
SkinThickness	20.54	23.00	15.95
Insulin	140.67	125.00	86.38
BMI	32.46	32.30	6.88
Diabetes pedigree function	0.47	0.37	0.33
Age	33.24	29.00	11.76

Visualization:

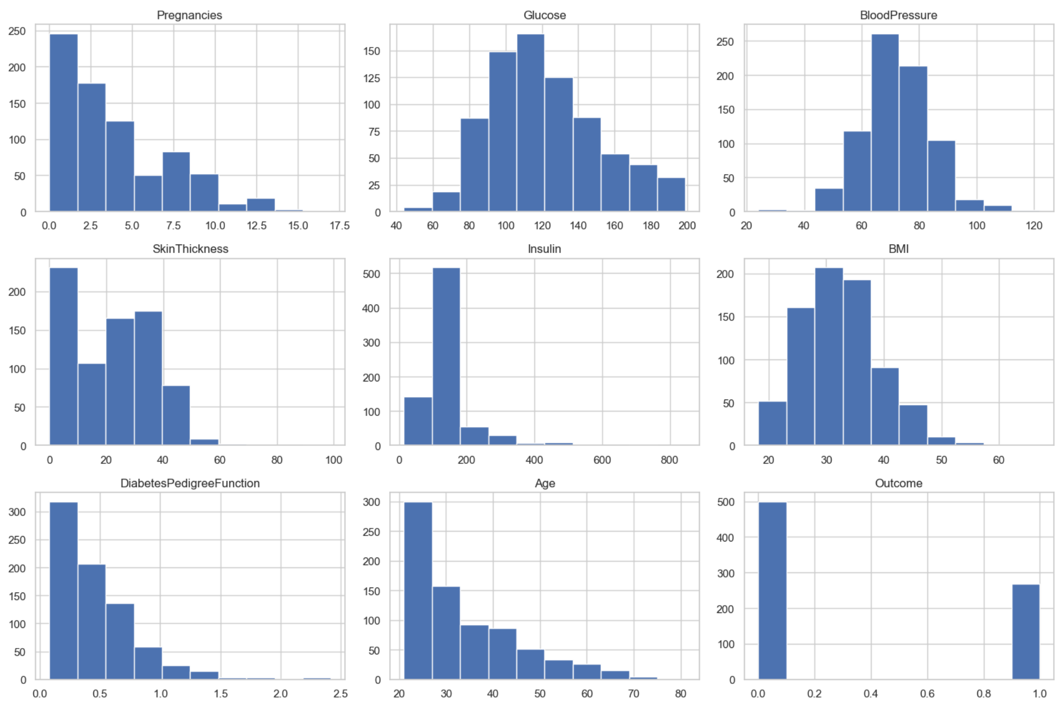


Figure 10 Show the distribution of numerical variables using histograms

By using histograms we can visualize the distribution of each variable. It helps to identify the skewed distributions and potential outliers.

Irregularities, Trends and patterns:

- The histograms reveal some useful trends and issues in the diabetes data. Pregnancies highly skewness towards right with most women experiencing 0 to 5 pregnancies and only few experiencing above 10 pregnancies.
- The glucose itself is nearly normal but skewed to the right with majority of the findings ranging between 100 and 140 mg/dL.
- BloodPressure is approximately normal, and that is approximately 70 mmHg.
- The body mass index is near normal with the majority falling under the 25-40 range implying that the majority of the population is overweight or obese.
- DiabetesPedigreeFunction is highly right skewed with majority of the population having low genetic risk scores.
- A number of variables are skewed towards right and the subjects of this dataset are predominantly young women who are overweight and obese with a few high glucose levels.

4.3 Descriptive Analysis

From the descriptive analysis the relationships among variables and diabetes outcomes were descriptively analyzed to reveal the main patterns and relationships.

```
df.groupby('Outcome').mean()
```

✓ 0.0s

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age
Outcome								
0	3.298000	110.682000	70.920000	19.664000	127.792000	30.885600	0.429734	31.190000
1	4.865672	142.130597	75.123134	22.164179	164.701493	35.383582	0.550500	37.067164

Figure 11 Mean values grouped by diabetes outcome

The mean shows the average value of variables by grouping those numerical variables by diabetes we can easily compare averages between people with and without diabetes.

	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunction	Age	Outcome
Pregnancies	1.000000	0.128213	0.208615	-0.081672	0.025047	0.021559	-0.033523	0.544341	0.221898
Glucose	0.128213	1.000000	0.218937	0.074455	0.419451	0.231049	0.137327	0.266909	0.492782
BloodPressure	0.208615	0.218937	1.000000	0.007937	0.045363	0.281257	-0.002378	0.324915	0.165723
SkinThickness	-0.081672	0.074455	0.007937	1.000000	0.183368	0.381109	0.183928	-0.113970	0.074752
Insulin	0.025047	0.419451	0.045363	0.183368	1.000000	0.180241	0.126503	0.097101	0.203790
BMI	0.021559	0.231049	0.281257	0.381109	0.180241	1.000000	0.153438	0.025597	0.312038
DiabetesPedigreeFunction	-0.033523	0.137327	-0.002378	0.183928	0.126503	0.153438	1.000000	0.033561	0.173844
Age	0.544341	0.266909	0.324915	-0.113970	0.097101	0.025597	0.033561	1.000000	0.238356
Outcome	0.221898	0.492782	0.165723	0.074752	0.203790	0.312038	0.173844	0.238356	1.000000

Figure 12 Correlation Matrix

Correlation matrix shows linear relationships between numerical variables. Positive correlation means both variables increase together. Negative correlation means one increases, the other decreases.

4.4 Central Tendency and Dispersion

Central tendency are used to identify the central or typical value of a dataset. They include a summary, which sows the overall behavior of numerical variables and helps to understand where most data values are concentrated.

Dispersions measures are used to define how much the data is spread or how varied the data is around the central value. Whereas central tendency represents a normal value, dispersion represents how much the data values differ from one another.

4.4.1 Central Tendency

```python
#Mean
mean_values = df.mean()
mean_values
```

✓ 0.0s Python

```
Pregnancies                   3.845052
Glucose                     121.656250
BloodPressure                72.386719
SkinThickness                20.536458
Insulin                     140.671875
BMI                          32.455208
DiabetesPedigreeFunction      0.471876
Age                          33.240885
Outcome                       0.348958
dtype: float64
```

Figure 13 Find the mean values

- Represent the average value of a variable. It provides a baseline comparison between diabetic and non diabetic groups

```
#Median
median_values = df.median()
median_values
```

✓ 0.0s Python

```
Pregnancies                   3.0000
Glucose                     117.0000
BloodPressure                72.0000
SkinThickness                23.0000
Insulin                     125.0000
BMI                          32.3000
DiabetesPedigreeFunction      0.3725
Age                          29.0000
Outcome                       0.0000
dtype: float64
```

Figure 14 Find the median

- Median represent the middle value of variables. It helps to detect skewed distributions. For example if median blood glucose is lower than mean, the data is right skewed.

4.4.2 Dispersion

```
#Variance
variance_values = df.var()
variance_values
```

✓ 0.0s Python

```
Pregnancies                   11.354056
Glucose                      926.489244
BloodPressure                146.328741
SkinThickness                254.473245
Insulin                     7462.033002
BMI                           47.268056
DiabetesPedigreeFunction       0.109779
Age                          138.303046
Outcome                        0.227483
dtype: float64
```

Figure 15 Variance

- Variance shows how the data spread out from the mean.

```python
#Standard Deviation
std_dev_values = df.std()
std_dev_values
```

✓ 0.0s Python

```
Pregnancies                  3.369578
Glucose                     30.438286
BloodPressure               12.096642
SkinThickness               15.952218
Insulin                     86.383060
BMI                          6.875177
DiabetesPedigreeFunction     0.331329
Age                         11.760232
Outcome                      0.476951
dtype: float64
```

Figure 16 Standard deviation

- Square root of variance represented through standard deviation.

Table 4 Summary Table

	Mean	Median	Variance	Std Deviation
Pregnancies	3.85	3.00	11.35	3.36
Glucose	121.66	117.00	926.48	30.44
BloodPressure	72.39	72.00	146.32	12.09
SkinThickness	20.54	23.00	254.47	15.95
Insulin	140.67	125.00	7462.03	86.38
BMI	32.45	32.30	47.26	6.87
DiabetesPedigreeFunction	0.471	0.37	0.10	0.33
Age	33.24	29.00	138.30	11.76

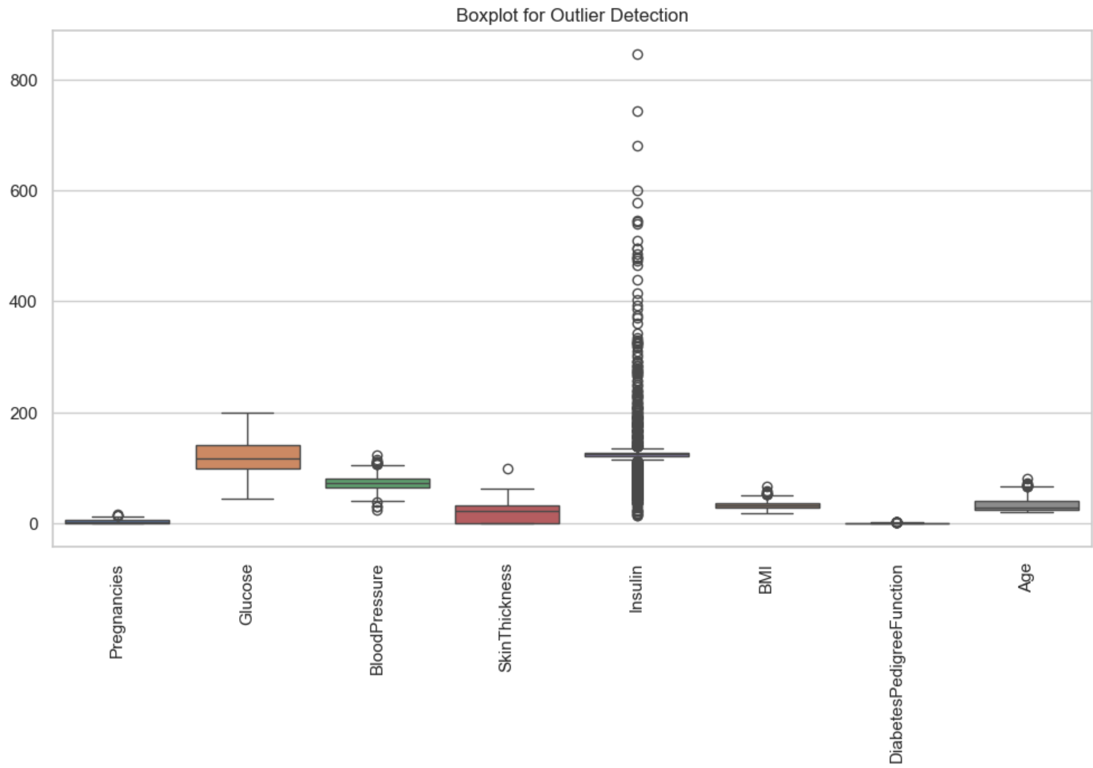


Figure 17 Detect outliers using boxplot

The dispersion and measures of central tendency give valuable information on the distribution and variability of the characteristics of the diabetes dataset. The average and the median of the majority of the variables are relatively close meaning that the data is quite balanced in the case of some of its characteristics like Blood pressure, BMI, and Diabetes pedigree function. Nevertheless, there are certain variables that are observed to have significant variations between the mean and median indicating the likelihood of skewness and occurrence of outliers.

An example is that the mean of Glucose is 121.66 and the median of Glucose is 117.00, and hence the right skewness is a slight skew, that is, there are some patients who have high glucose readings that increase the average. On the same note, Insulin is characterized by a large difference between mean (140.67) and median (125.00), a very high variance (7462.03) and standard deviation(86.38) meaning that there is a great variability and that there may be extreme values of insulin. Other variables that display a little skewness are the Age variable that has a mean of 33.24 and a median of 29.00 indicating that older patients would contribute to the average.

The values of standard deviation indicate the extent of dispersion of the data. The variables like Glucose (30.44), Insulin (86.38), and Age (11.76) are relatively high, indicating that the measuring of patients varies across a high range. Conversely, Diabetes pedigree function has lower standard deviation (0.33), which means that the values of the patients are more constant.

5. Data Visualization

5.1 Histogram

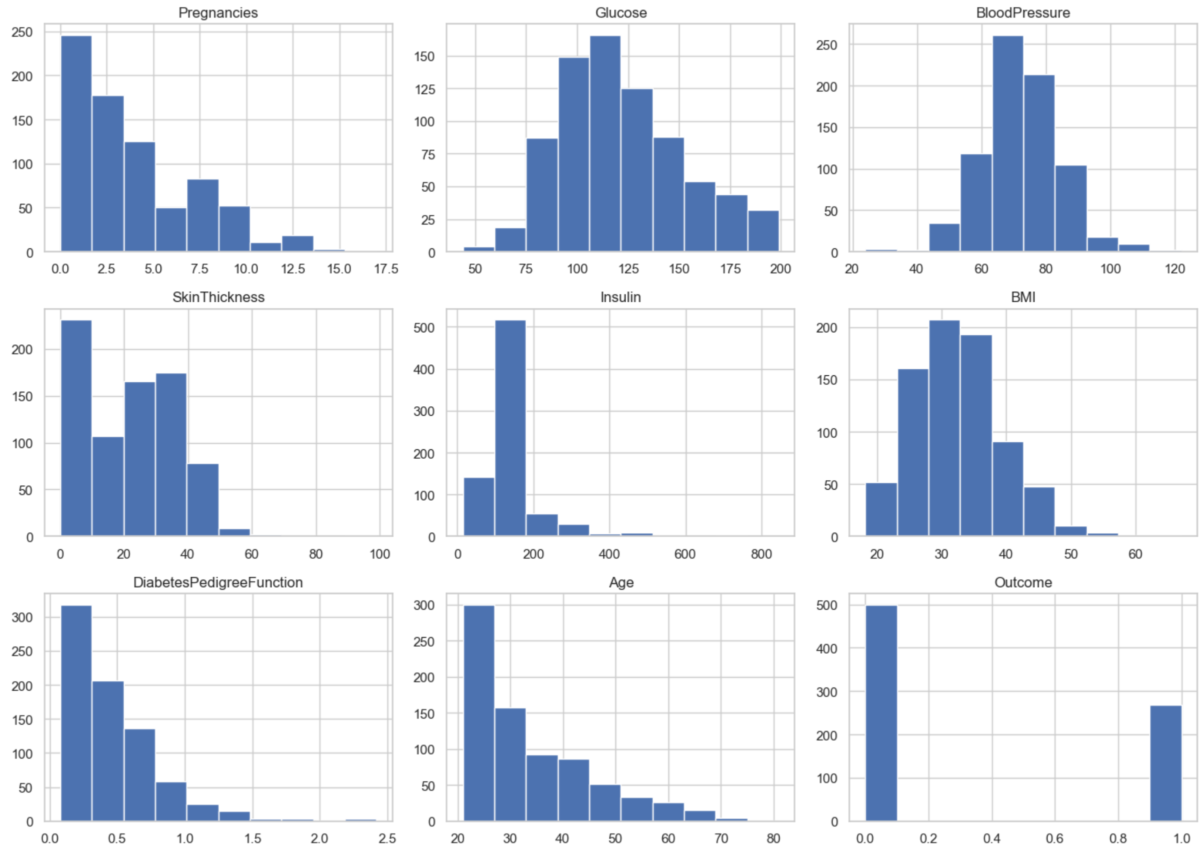


Figure 18 Histogram

The histograms give a very clear picture that represents the distribution of each attribute in the diabetes dataset in the patient population. The Pregnancies attribute displays a skewed distribution with a majority of the women bearing 0 to 5 pregnancies meaning that the higher number of pregnancies is less prevalent in the data.

The spread of the values in the Glucose attribute is large, and the concentration between 100 and 150 mg/dL is prominent. It implies that a significant part of the population is close to the prediabetes and diabetes threshold, which shows that glucose is a highly important risk factor.

The BMI histogram indicates that the majority of individuals range between 20 to 40 kg/m² in the BMI, which implies that there is a dominance of overweight and obese people. Because obesity is a recognized risk factor of diabetes, such distribution can justify the applicability of BMI in the prediction of diabetes.

The skewed distribution of the Insulin attribute is that there are numerous low values of the Insulin attribute and a small number of extremely high values. This implies that the insulin levels vary significantly and it could be as a result of insulin resistance or abnormal secretion of insulin in diabetics.

Diabetes Pedigree Function is highly clustered at the lower end and this implies that genetic predisposition differs across individuals, however most patients in the dataset have a low genetic predisposition.

All in all, histograms are useful in determining skewness, spread, and abnormal trends in data that are critical in determining the correct preprocessing methods and machine learning models.

5.2 Boxplot

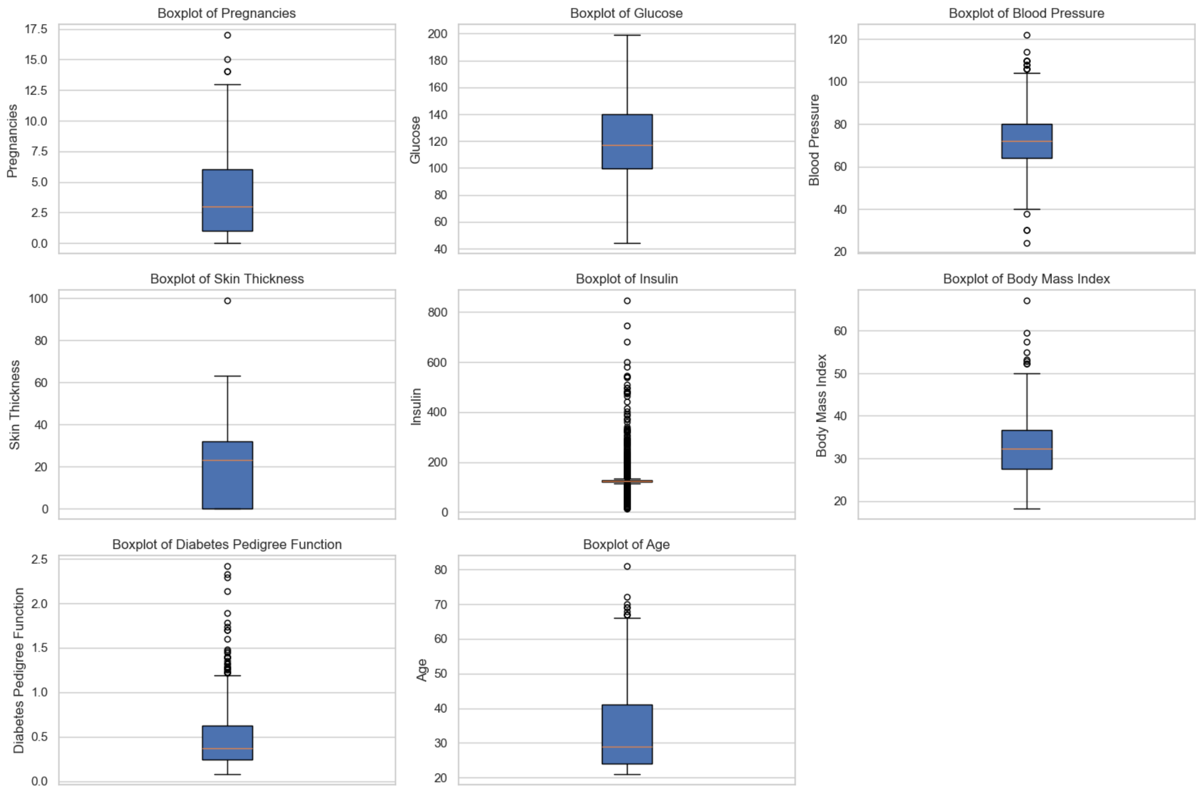


Figure 19 Boxplot

Box plots were used to determine the distribution and occurrence of outliers among the main medical characteristics, including glucose, blood pressure, BMI, and insulin. Every box plot shows the median, the interquartile range (IQR), and the values that are not within 1.5 times the IQR which can be discussed as the possible outliers.

The Glucose boxplot shows that there are a few extreme values which are above the upper quartile, which shows the patients who have unusual high glucose levels. Such cases are clinically relevant, since they can be severe diabetic conditions.

The Insulin attribute presents the most outliers, and this attests to the fact that it has a large variance in the dataset.

The BMI boxplot shows that dispersion is moderate, and there are some outliers in the upper part, which implies the occurrence of severely obese people. This supports the BMI as a significant predictor as well as indicates the necessity of dealing with outliers.

Box plots are very instrumental in identifying any abnormal data and making decisions on how to treat the outliers, as it helps make sure that the predictive models are healthy and sound.

5.3 Scatter Plot

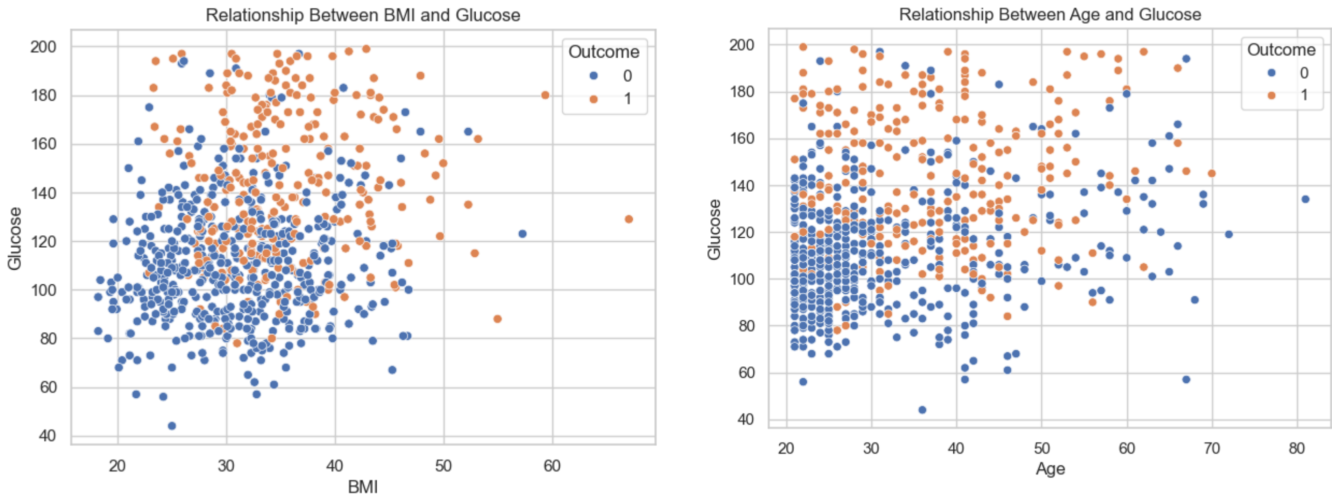


Figure 20 Scatter plots

The relationships between the main risk factors and the results of diabetes were analyzed using scatter plots. The BMI and Glucose scatter plot indicates that the values of BMI are positively related to the amount of glucose, whereby high BMI will be associated with high levels of glucose. This cluster implies that obesity is one of the factors that predispose high blood glucose, which strengthens its health hazard as a risk factor of diabetes.

As can be seen in the Ageand Glucose scatter plot, older patients have higher glucose more often, but not all old patients have low glucose levels. This shows that age is also a causative factor but the risk of diabetes is not confined to the older generations.

The integration of diabetes Outcome into the scatter plot with the help of color differentiation will create distinct clusters in which patients with diabetes are likely to be characterized by higher glucose and BMI rates than those who are not diabetic. These visualizations support relationships obtained with descriptive statistics and correlation test.

Scatter plots are considered to be necessary in order to prove correlations visually, identify clusters, and promote the derivation of hypotheses on diabetes risk factors.

5.4 Correlation Matrix

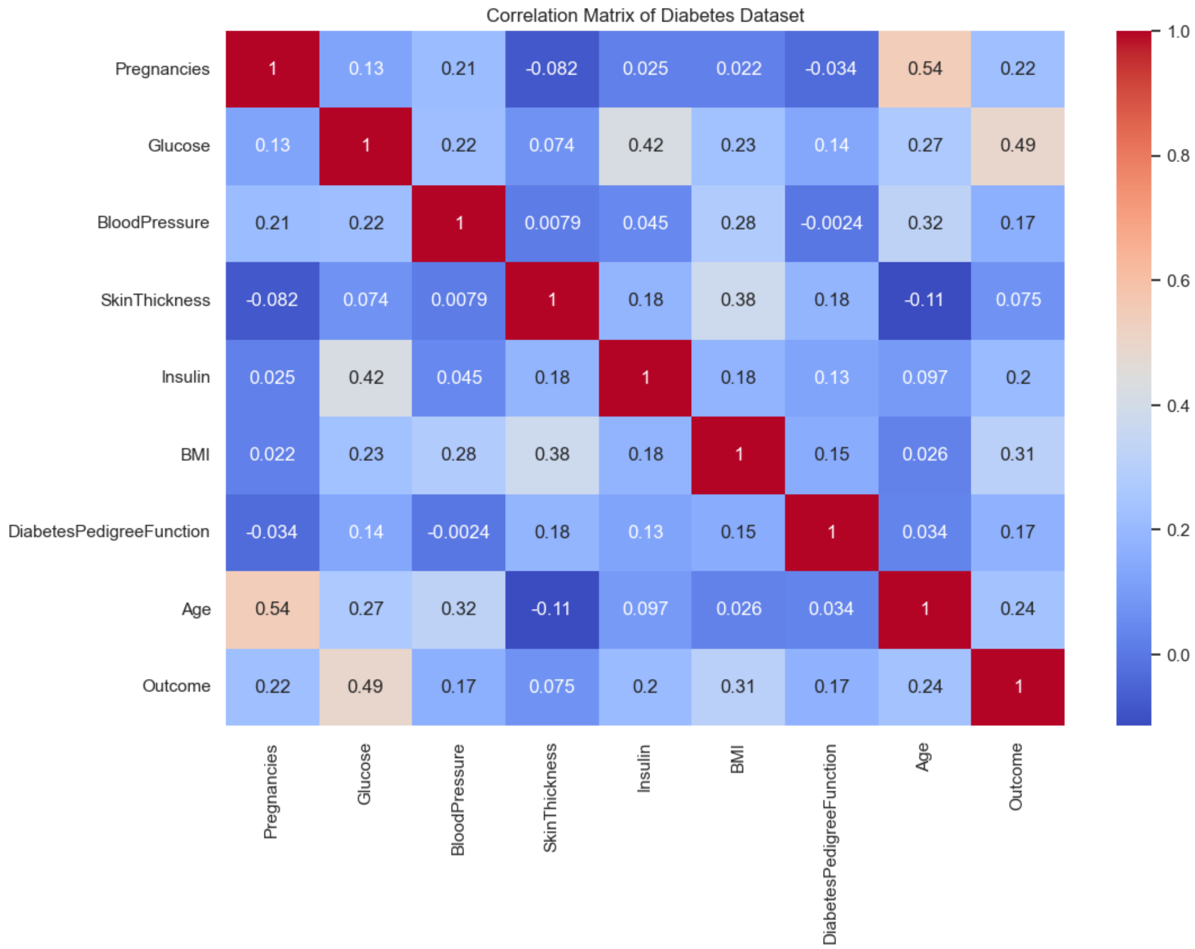


Figure 21 Correlation Matrix

The correlation heat map gives the overall view of the linear relations between all the numerical attributes in the diabetes dataset. The intensity of color shows how intense the correlation is and darker colors show stronger associations.

The positive correlation of the Glucose attribute with the diabetes outcome is the highest, which proves that it is the most powerful predictor of diabetes. This observation is in line with the science of medicine because high levels of blood glucose are directly linked with the diagnosis of diabetes.

The outcome variable has moderate positive correlations with both BMI and Age, indicating that obesity and aging have a high tendency to predispose one to diabetes. Conversely, other attributes like skin thickness and diabetes pedigree functioning exhibit weak correlations, which mean that it does not affect diabetes outcomes directly.

The heat map is also useful in highlighting multicollinearity among features, useful in determining the right machine learning models to use and also in removing redundant predictors.

In general, the correlation matrix is an important instrument to select the features and have a statistical rationale to the model-building decisions in the subsequent analysis phase.

6. Machine Learning Models and Performance Evaluation

6.1 Data Preparation for Modeling

Define Features and Target

```
#Define Features (X) and Target (y)
X = df.drop("Outcome", axis=1)
y = df["Outcome"]
```
✓ 0.0s Python

Figure 22 Define features and targets of the datset

In machine learning, the data needs to be broken into input variables (features) and an output variable (target) and then a model is trained.

Features (X) are the independent variables with which the model makes predictions. Under X, the health related measures involved in this dataset are Glucose, BMI, Age, Blood Pressure and other clinical characteristics.

Target (y) will be the dependent variable to be predicted by the model. Here, the target variable is Outcome that presents the value of whether an individual is diabetic (1) or not (0).

This division is required since machine learning models are trained to learn patterns by examining the relationship of the features (X) to the target (y). The model would have no way of knowing what data it will learn and what outcome it will predict without having a clear definition of both X and y.

Split the Dataset

```
#Split the Dataset
X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42, stratify=y
)
```
✓ 0.0s Python

Figure 23 Split the dataset

The data is divided into training and testing to train a model and test its performance on the data it has not seen. Evaluation is done on a test size of 20% and 80% of the data is to be used in training. Distribution of classes makes sure that the distribution of classes is preserved which produces a fair and reliable performance measurement.

Feature Scaling

```
#Feature scaling
scaler = StandardScaler()

X_train_scaled = scaler.fit_transform(X_train)
X_test_scaled = scaler.transform(X_test)
```
✓ 0.0s Python

Figure 24 Feature scaling

The feature scaling makes all the input features have an equal contribution to the model. Logistic Regression is sensitive to scopes of features and thus standardising them enhances the performance of the model and speed of training.

To bring the features to a mean of 0 and standard deviation of 1 the features are rescaled using StandardScaler.

This is necessary to avoid dominance of features with bigger ranges in the model and scaling parity between training and test data.

6.2 Model Selection

Choose two models: Random Forest and Logistic Regression.

Logistic Regression: It was selected due to its simplicity and high interpretability in binary classification problems such as diabetes prediction. It gives out coefficients to comprehend the effects of features, and therefore it is best in describing relationships in a health dataset. It makes the assumption of linear relationships and is computationally efficient with this moderate-sized dataset (768 samples), and interpretability over possible accuracy in complex environments.

Random Forest: Chosen because it is the most suitable ensemble model that can deal with non-linear relationships and interactions of features, and avoid over fitting by averaging out the results of numerous decision trees. It is more accurate on unbalanced data such as this (~65% non-diabetic) and gives importance rankings on the features. This sacrifices some interpretability to better predictive performance, appropriate in this case due to the possible complexities of the dataset (ex. interactions between BMI and Age).

This choice provides the characteristics of the dataset with a balance: small size, possible imbalances, and their necessity to provide both explainable information (Logistic) and high accuracy (Random Forest).

6.3 Model Training

Training was done with the training dataset on each model: The Logistic Regression was trained using scaled features to acquire a linear relationship between predictors and the diabetes outcome. Random Forest was trained based on several decision trees and this enabled the model to form complicated interactions between features. Training allows the models to acquire the patterns based on historical patient data which could be applied to the unseen cases of diabetes status prediction.

```python
# Train models
# 1. Logistic Regression
logreg = LogisticRegression(random_state=42)
logreg.fit(X_train_scaled, y_train)
```

✓ 0.0s Python

```
LogisticRegression
▸ Parameters
```

```python
# 2. Random Forest
rf = RandomForestClassifier(n_estimators=100, random_state=42)
rf.fit(X_train, y_train)
```

✓ 0.2s Python

```
RandomForestClassifier
▸ Parameters
```

Figure 25 Training the model

Make Predictions

```python
#Make Predictions
# Logistic Regression
y_pred_logreg = logreg.predict(X_test_scaled)
```

✓ 0.0s Python

```python
# Random Forest Predictions
y_pred_rf = rf.predict(X_test)
```

✓ 0.0s Python

Figure 26 Make Predictions

6.4 Performance Evaluation

6.4.1 Evaluate Logistics Regression

```
--- Logistic Regression ---
Accuracy: 0.7142857142857143
Precision: 0.6136363636363636
Recall: 0.5
F1 Score: 0.5510204081632653
```

Figure 27 Evaluate logistic regression

Accuracy (71.43%) - Accuracy is a ratio of the rightly classified ones among the overall number of predictions. The accuracy of 71.43 means that this model was able to predict diabetes outcomes with the necessary accuracy of about 7 out of 10 patients. This implies that there is relatively good overall performance, albeit there exist some room to improve.

Precision (61.36%) - Precision refers to the number of the patients who were predicted to be diabetic but were actually diabetic. This shows a precision value of 61.36 percent implying that when the model predicts that a patient is diabetic, it is correct 61.36 percent of the time. This shows that there are some false positive predictions, that is, patients who are not diabetic were wrongly diagnosed as diabetic.

Recall (50%) - Recall is used to determine the capability of the model to correctly identify real diabetic patients. Recall score of 50 percent indicates that the model was only able to identify half of the actual cases of diabetes. This implies that the model does not capture numerous cases of diabetic patients (false negatives) which is a serious drawback when making a medical diagnosis.

F1 Score (55.10%) - F1 score is a harmonic mean of precision and recall that gives an equal measure of the performance of the model. The F1 score is 55.10% which means that there is a moderate proportion between the correct rate of identifying diabetic patients and false rate of classifying them. The fact that the F1 score is relatively low is indicative of the poor recall by the model.

Confusion Matrix

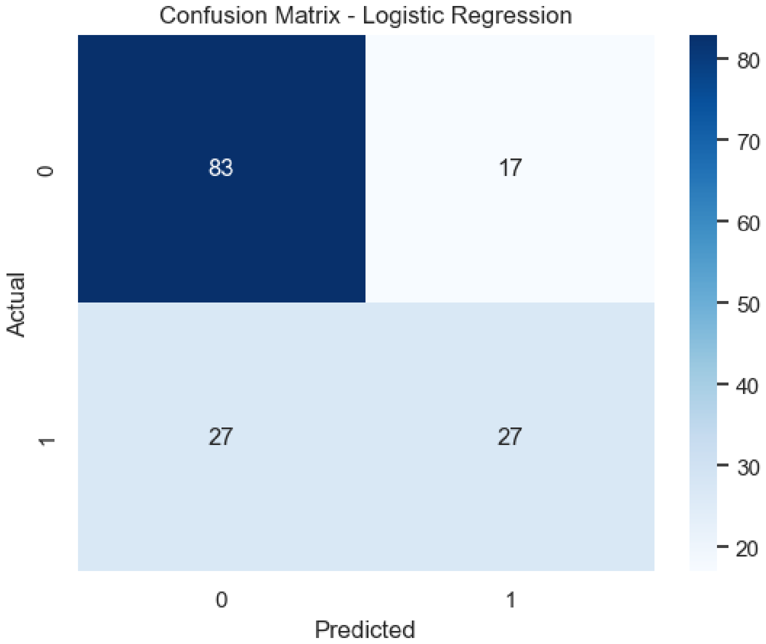


Figure 28 Confusion Matrix (Logistic Regression)

Classification Report

```
Classification Report:
              precision    recall  f1-score   support

           0       0.75      0.83      0.79       100
           1       0.61      0.50      0.55        54

    accuracy                           0.71       154
   macro avg       0.68      0.67      0.67       154
weighted avg       0.71      0.71      0.71       154
```

Figure 29 Classification report (logistic regression)

Class 0 (Non-Diabetic)

- Precision: 0.75 - This implies that 75% of the cases that were forecasted to be non-diabetic were correctly categorized. There are few false positive errors of this class in the model.
- Recall: 0.83 - Recall rate of 83% means that the model has been able to identify the majority of the real non-diabetic cases.
- F1-score: 0.79 - The high F1-score represents a good balance of objectivity and recall of class 0.
- Support: 100 - The test sample had 100 actual non-diabetic samples.

Class 1 (Diabetic)

- Precision: 0.61 - This shows that 61% of the cases that had been classified as diabetic were really diabetic that is, a few non-diabetic cases were wrongly classified as diabetic.
- Recall: 0.50 - A recall of 50% indicates that the model was able to recognize half of the true cases of diabetes missing the other half.
- F1-score: 0.55 - The reduced F1-score implies poor performance with respect to the identification of diabetic patients when compared to non-diabetic patients.
- Support: 54 - The test data had 54 real diabetic samples.

6.4.2 Evaluate Random Forest

```
--- Random Forest ---
Accuracy: 0.7532467532467533
Precision: 0.6739130434782609
Recall: 0.5740740740740741
F1 Score: 0.62
```

Figure 30 Evaluate Random forest

Accuracy (75.3%) - The 0.753 accuracy value shows that the model of the Random Forest is rightly classifying about 75.3% of the total cases in the test set. This demonstrates that the model is quite good in general in making the distinction between diabetic and non-diabetic patients.

Precision (67.4%) - The value of precision is 0.674 which states that 67.4% out of all the instances predicted to be diabetic were actually diabetic. This is an indication that the model is average in preventing false positive outcomes and thus minimizes the false diagnoses of non-diabetic patients.

Recall (57.4%) - The recall score of 0.574 shows that the model was able to correctly identify 57.4% of all the real diabetic cases. This means that there was a significant proportion of diabetic patients who were not identified using the model which can be disadvantageous in situations where positive cases are decisive of missing.

F1 Score (62%) - F1 score of 0.62 is an indication of a trade off between accuracy and recall. As it is neither too high nor too low, it means that the model has a moderate trade-off between false alarms and the identification of diabetic patients.

Confusion Matrix

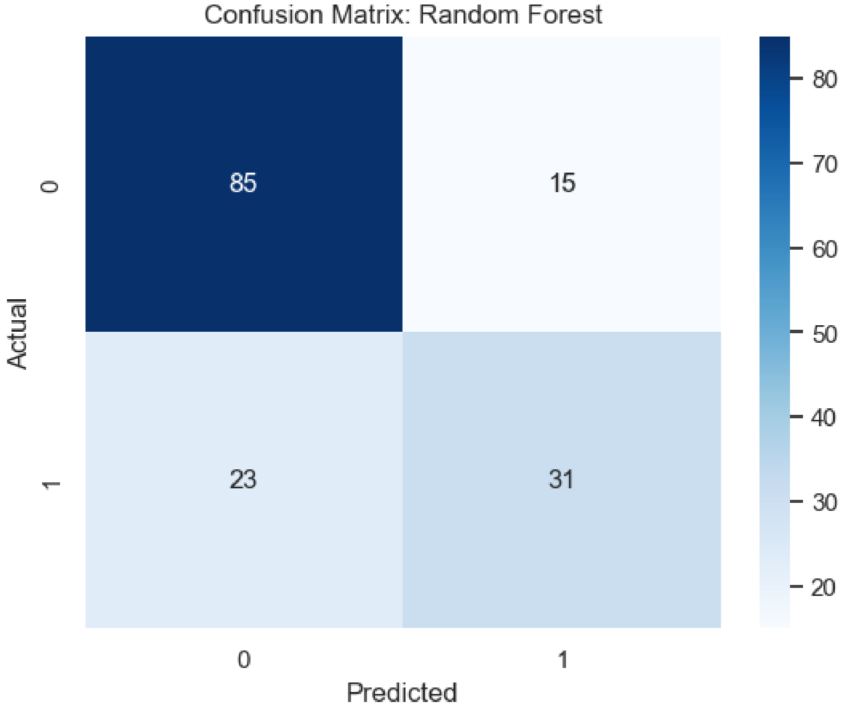


Figure 31 Confusion matrix (Random forest)

Classification Report

Classification Report:

	precision	recall	f1-score	support
0	0.79	0.85	0.82	100
1	0.67	0.57	0.62	54
accuracy			0.75	154
macro avg	0.73	0.71	0.72	154
weighted avg	0.75	0.75	0.75	154

Figure 32 Classification Report (Random forest)

Class 0 (Non-Diabetic)

- Precision: 0.79 - This implies that 79% of the cases that were forecasted to be not diabetic were correctly identified, which means that the model does not give a false positive with a high error rate in this category.
- Recall: 0.85 - The recall rate was 85% and this indicates that most of the true non-diabetic instances were identified by the model.
- F1-Score: 0.82 - The balance between the precision and the recall is high as seen by the high F1-score which demonstrates good performance of class 0.

- Support: 100 - The test data had 100 real non-diabetic cases.

Class 1 (Diabetic)

- Precision: 0.67 - This implies that 67% of the cases that were predicted to be diabetic actually were diabetic thus it is false positive.
- Recall: 0.57 - A recall of 57% demonstrates that the model accurately identified a little more than half of the real cases of diabetes, and the other ones were not detected.
- F1-Score: 0.62 - The F1-score shows the moderate performance in the detection of diabetic patients.
- Support: 54 - The test dataset had 54 real diabetic cases.

6.5 Model Comparison

In this section, the performance of the two selected machine learning models—Logistic Regression and Random Forest is compared based on key evaluation metrics.

	Model	Accuracy	Precision	Recall	F1 Score
0	Logistic Regression	0.714286	0.613636	0.500000	0.55102
1	Random Forest	0.753247	0.673913	0.574074	0.62000

Figure 33 Model comparison

The Linear model, Logistic Regression, had a precision of 61.4% as well as accuracy of about 71.4% and a recall of 50% and a F1 score of 55.1%. This model works well in case of a binary classification problem such as the prediction of diabetes where the answer is either diabetic (1) or non-diabetic (0). Its advantages consist in the fact that it is very easy and interpretable since coefficients directly can be analyzed to determine what features like glucose levels or BMI have an influence on predictions. Nevertheless, it has a less impressive recall meaning that it lacks a significant proportion of real diabetic cases, and this may be important in a medical context where detection of people at risk is a priority.

In contrast Random forest model, an ensemble algorithm combine multiple decision trees, had a higher score across all measures 75.3%, precision 67.4%, recall 57.4%, and F1 score 62% in comparison to the Logistic Regression. This enhancement is an indication that the Random Forest is more effective in capturing non-linear interactions and relationships among the features in the dataset like the overall effect of age, insulin and diabetes pedigree function. Although it provides better overall performance, it is not as interpretable as Logistic Regression because it is a black-box model, but some clue can be given by ranking the importance of the features.

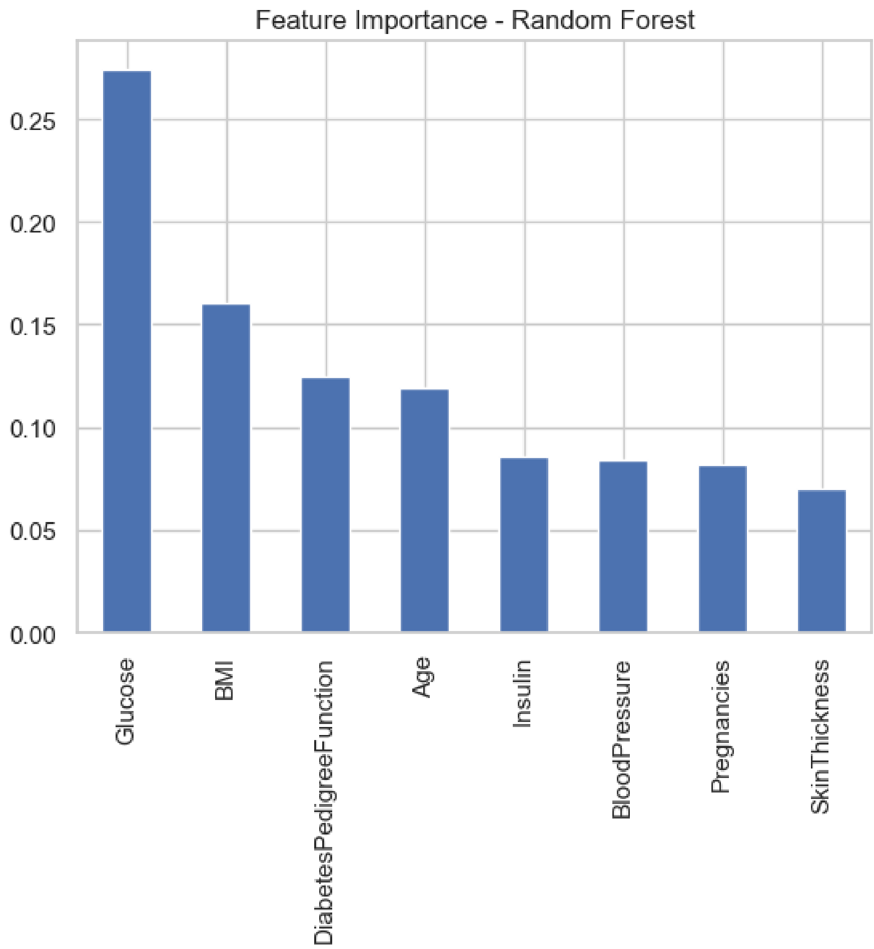
Based on the evaluation metrics, the best model to predict diabetes is Random Forest since it is more balanced in refusing false positives and false negative represented by the higher F1 score. Despite the fact that Logistic Regression is easier and more understandable, the difference in the performance warrants the choice of the Random Forest, particularly due to the possible complexity of patterns in the dataset. In any clinical use, where accuracy and recall are the most important to reduce the number of missed diagnoses, the benefits of Random Forest outweigh the issue of interpretability. These results can be improved further in the future by hyperparameter optimization or by using more models.

7. Interpretation, Conclusion and References.

7.1 Summary of Findings

This project has evaluated the diabetes data set to determine significant risk factors and to construct predictive machine learning models. The results are outlined below in relation to the research questions.

1. What are the greatest clinical risk factors related to diabetes?



The bar chart shows the feature importance based on the random forest model. The scores of feature importance are brought to 1.0, which reflects the proportion of each variable to the predictive strength of the model in defining the results of diabetes (0: no diabetes, 1: diabetes). The taller bars represent aspects that have a more significant impact on the decisions made by the model that can be used to determine the most important risk factors to relate to the prevalence of diabetes.

- Glucose - This becomes the most significant feature and it is in accordance with the clinical knowledge that the high levels of glucose in plasma are the leading indicators of diabetes. This variable is very important in dividing decisions in the model indicating that it is a powerful predictor of the risk of diseases.
- BMI - In the second position, there is Body Mass Index, which shows obesity as one of the major modifiable risk factors. An increase in the BMI levels is associated with a rise in insulin resistance, which is a typical antecedent of type 2 diabetes.
- Diabetes pedigree Function – Diabetes pedigree function is a genetic risk model that quantifies the family history of diabetes and emphasizes the hereditary nature of the disease. It gives information about non-controllable determinants on prevalence.
- Age - Age has a moderate level of importance, which aligns with the literature stating that the risk of diabetes grows with the growing age as a result of such factors as declining physical activity and metabolic changes.
- Insulin - Serum insulin levels play a less significant, but significant role, which is related to the contribution of insulin deregulation to the pathophysiology of diabetes.
- Blood pressure, Pregnancies and Skin thickness - These variables show lower importance, which implies that they are introduced as secondary predictors in this dataset. As an example, increased blood pressure can be a symptom of comorbid conditions such as hypertension, and pregnancies are associated with the risk of gestational diabetes among women, whereas skin thickness can be used as a proxy measure of the body fat distribution.

2. Does Age correlate significantly with diabetes status?

- Yes. The findings indicated that there was a positive relationship between Age and diabetes. When age is increase, chances of getting diabetes also increase. The prevalence of diabetes among older people in the dataset was more than that of the younger ones. The correlation was however moderate and not as high as glucose.

3. What is the strength of the relationship between Glucose and diabetes prediction?

- Among all the variables, glucose show the strongest correlation with diabetes. An augmented glucose level was also a strong indication of the likelihood to be diagnosed as diabetic. Glucose had the greatest effect on prediction as demonstrated before. Glucose was the most significant feature in tree-based models. This affirms that the level of blood glucose is the most valuable predictor of diabetes in this dataset.

4. Does an increase in BMI lead to a high probability of diabetes?

- Yes. BMI is positively correlated with diabetes. A higher value in the BMI of the patients increased their chances of diabetes. This implies that obesity and overweight are risk factors. However BMI is not sufficient to predict diabetes. It is more effective in conjunction with age and glucose.

5. Does DiabetesPedigreeFunction increase the prevalence of diabetes?

- Yes. DiabetesPedigreeFunction was positively correlated to diabetes. Those who had diabetes were more prone to do so due to a high hereditary risk score. The finding indicates the fact that genetics is vital in the development of diabetes.

6. What machine learning model gives the most effective performance?

- Logistic Regression and Random Forest were tested. The good interpretability and reasonable accuracy were offered by the Logistic Regression but the Random Forest was more accurate, precise and had better F1-score. Therefore, Random Forest was better in the general prediction of diabetes. But the Logistic Regression is simpler to interpret and explain which is essential in the medical decision-making.

7.2 Discuss of Implications

The results of this research are significant in terms of their implication on healthcare and prevention practice.

1. Monitoring of Glucose is essential - As glucose is the best predictor, periodic blood glucose screening is recommended particularly among the high-risk people.
2. Obesity Prevention Programs.

   Positive correlation between BMI and diabetes demonstrates the significance of:

   - Encouraging physical exercise.
   - Encouraging healthy diets
   - Reducing obesity levels

3. Target the Older Populations - As diabetes risks are linked to age, health screening programs should target more on those who are middle aged and older.
4. Family History Awareness - Increased risk is to be taught to individuals whose family history has shown diabetes and preventive lifestyles should be instilled among them early in life.

These lessons can guide policymakers to come up with more effective public health initiatives to reduce the prevalence of diabetes.

7.3 Limitations of the study

Despite the fact that the study includes valuable findings, it still has certain limitations:

- Limited Dataset Size: The samples used in the dataset are few and this can influence the generalisation of the models.
- Missing or Zero Values: There were zero values of some variables (BMI or Insulin) which could be missing data and this can affect the accuracy.
- Lack of Lifestyle Variables: The data fail to capture some of the key lifestyle variables like diet, smoking, drinking, or exercise.
- Single Population Dataset: The data is a reflection of a given population set and hence the finding might not be relevant to other ethnic or geographical sets.
- Model Limitations: Logistic Regression presupposes a linear relationship, Random Forest may overfit if not properly tuned.

Due to these limitations, the findings are to be taken carefully.

7.4 Future research directions

This research can be enhanced in the following ways in future studies:

1. Use Larger and More Diverse Datasets: More reliable data would be provided by including the data from other countries and ethnic groups.
2. Include More Variable: Include lifestyle factors like:
    - Physical activity
    - Diet
    - Smoking habits
    - Blood pressure history
3. Test More Advanced Models like:
    - Support Vector Machines (SVM)
    - Neural Networks

- Gradient Boosting (XGBoost)

4. Longitudinal Analysis - Future studies will be able to utilize the long term patient data instead of cross-sectional data in order to investigate the nature of diabetes development with time progression.
5. Explainable AI Techniques - Interpretability of complex models could be enhanced with SHAP or LIME application.

7.5 Conclusion

In conclusion, this study has determined significant risk factors related to diabetes through machine learning techniques.

The results show that:

- The most significant predictor of diabetes is glucose.
- BMI, Age and hereditary risk also play a significant role.
- Machine learning models, in particular, Random Forest, have the potential to predict diabetes.

The research findings confirm the significance of screening at an early age, encouraging healthy lifestyles, and special interventions on high risk individuals.

Machine learning is an effective solution in enhancing early diagnosis and aiding healthy judgment.

References and Citations

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